

Cloud Native Microservices and DevOps

Microservices Overview

Cloud-native microservices architectures decompose applications into independently deployable services communicating over lightweight protocols such as REST or gRPC. Container orchestration platforms like Kubernetes automate deployment, scaling, and failover operations.

CI/CD Pipelines

Modern CI/CD pipelines integrate GitHub Actions, Jenkins, ArgoCD, and Terraform for infrastructure automation. Pipelines perform linting, unit testing, container scanning, integration testing, and progressive deployments using canary or blue-green strategies.

Observability

Observability platforms aggregate metrics, traces, and logs using technologies such as Prometheus, Grafana, OpenTelemetry, and ELK Stack. Distributed tracing enables engineers to analyze latency propagation across interconnected services.

Scalability and Reliability

Reliability engineering practices include circuit breakers, retry policies, autoscaling, distributed caching, and chaos engineering. Teams use service meshes like Istio to manage traffic shaping, security policies, and telemetry collection.

Component	Purpose	Technology Example
Embedding Model	Vector Representation	SentenceTransformers
Vector Database	Semantic Retrieval	ChromaDB / FAISS
LLM	Answer Generation	Claude / GPT / Titan
Orchestration	Workflow Management	LangChain